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MARTINI SPORTSWEAR AROUCA BACKPACK

Proposed Improvements

[1]

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Require ments

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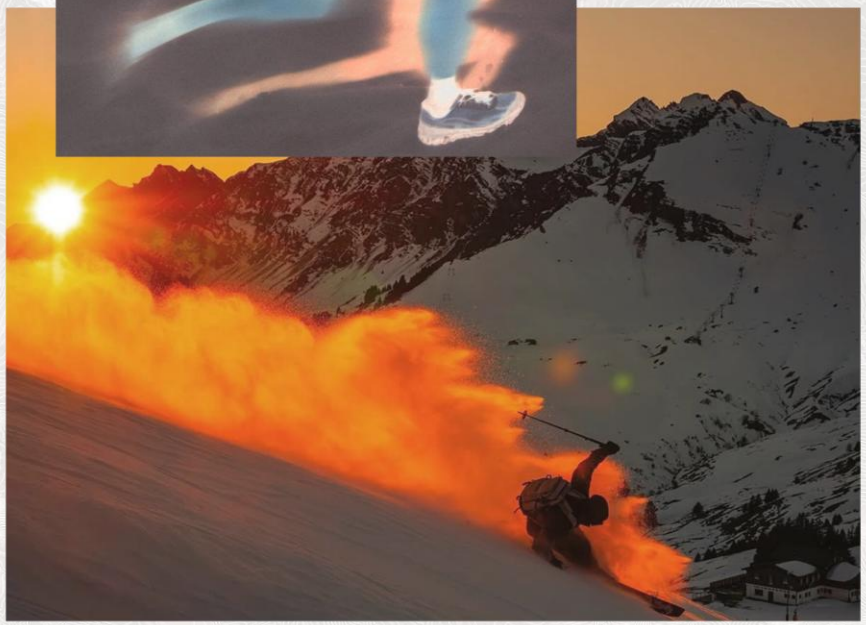
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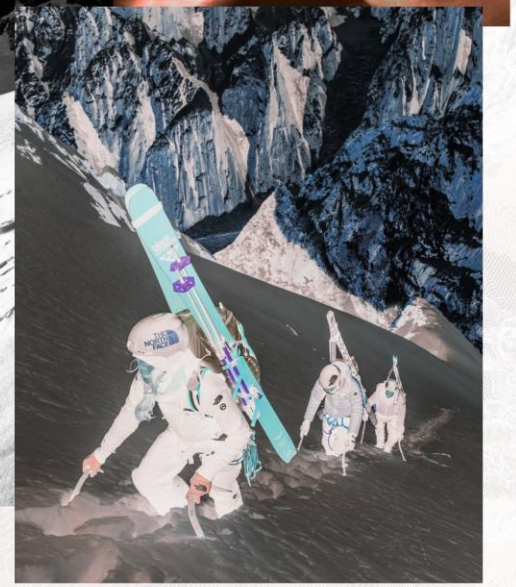












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Market Analysis

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[14] [15]



[13]

Dynafit Speed 24 / 22 W

- Different shoulder strap spacing for unisex and W version
- 755g for unisex, 700g for W version

Mammut
Aenergy ST 20-25

- Trail running style straps
- 740g

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[4]

DbJourney
Weigh Lighter 25L
Backpack

- Ultralight construction
- A-Style Ski-Mounts
- 470g



[7]

USWE
Carve 25L
Winter Backpack

- Racing-Buckle-Style Straps
- A-Style Ski-Mounts
- 1495g

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1st Concept

Material Substitution

Weight reduction
Improved wear comfort
Higher durability
Recyclability



[4]

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ALUULA Graflyte V-98	Martini Arouca backpack ripstop
polyethylene (PE), UHMWPE*	nylon-polyester-blend (PA-PES)
3 layers, two PE-films, woven UHMWPE core	2-3 layers, woven ripstop core, PA-PES-films, glued
98g/m² (30-49% reduction)	140-200 g/m²
1800 cycles (40-800% improvement)	200-1300 cycles
158 lb tear strength (5-690% improvement)	20-150 lb tear strength
waterproof	waterproof
heat-bonded/-welded seams	sewn seams
homogenous chemical composition -> recyclable	material blend -> not recyclable

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*ultra high molecular weight polyethylene

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2nd Concept

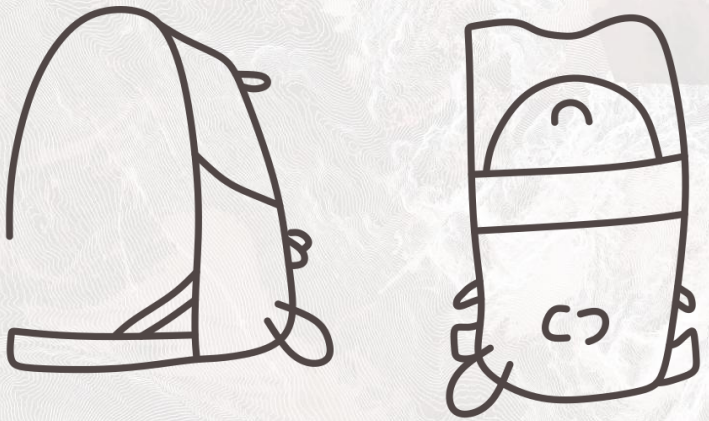
Dimension adjustments

Slimmer and taller shape

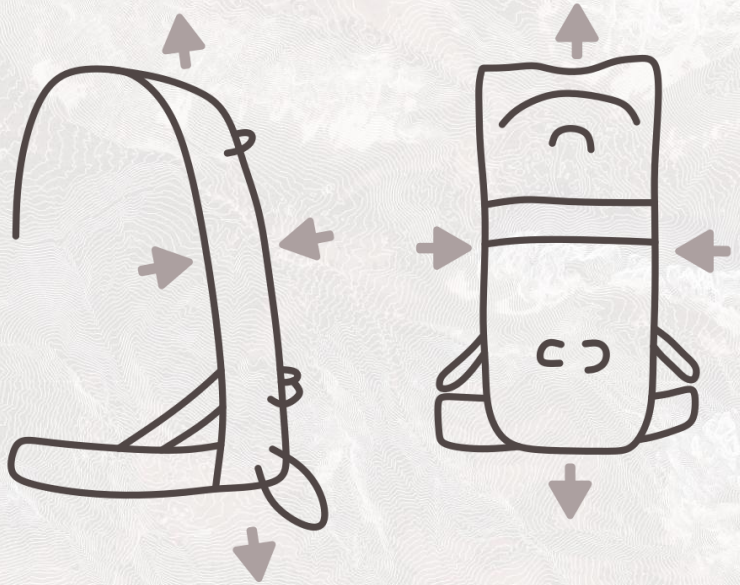
Same volume

Higher stability

Improved ergonomics



Before



After

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Study

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- 12 male, 8 female college-aged, healthy students
- 4 load conditions → backpack with 0, 5, 10, 15 % of bodyweight
- 2 placements → thoracic and lumbar
- Data capture → EMG, kinematics, 6 camera vicon system, subjective score and heart rate

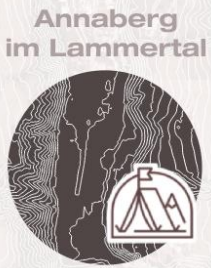
(Source: Devroey et al., 2007).

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Study – Facts



Result	Improvement
>= 10% of BW → significant forward tilt of chest (standing & walking)	slimmer and taller shape
The higher the load → activity of lumbar back extensors decreases significant	better load distribution across the spine
14 of 20 prefer lumbal position of backpack	

(Source: Devroey et al., 2007).



Mechanics

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Reduction of

- Lever arm
- Moment of inertia
- Lateral movements





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3rd Concept

USWE – No Dancing Monkey

Inspired by racing-seatbelts

Buckle & strap design

improves stability

additional storage space

[11]



[7]

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Why collaborate?

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Positively affect consumer's novelty perception (Cai et al., 2023) [9]

Expand market towards more customers

Potential for custom product development



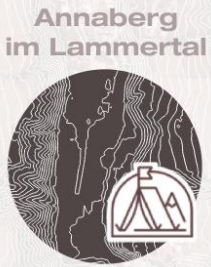
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Evaluation & Selection



criteria	factor	Martini Arouca Backpack	Concept 1 Material changes	Concept 2 Proportion changes	Concept 3 USWE NDM	Ensemble
Sustainability	0,1	6	9	6	6	9
Wear comfort	0,25	5	6	9	9	10
Total weight	0,25	5	9	5	4	8
Stability	0,2	5	5	9	9	10
Weather proof	0,1	8	10	8	8	10
Expected Life span	0,1	8	9	8	8	9
Total		37	48	45	44	56
Weighted total		5,7	7,55	7,5	7,25	9,3



Performance Testing

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1st Concept

Material Substitution

Loaded Bag Durability Test (Short- vs. Long-term)

2nd Concept

Dimension Adjustments

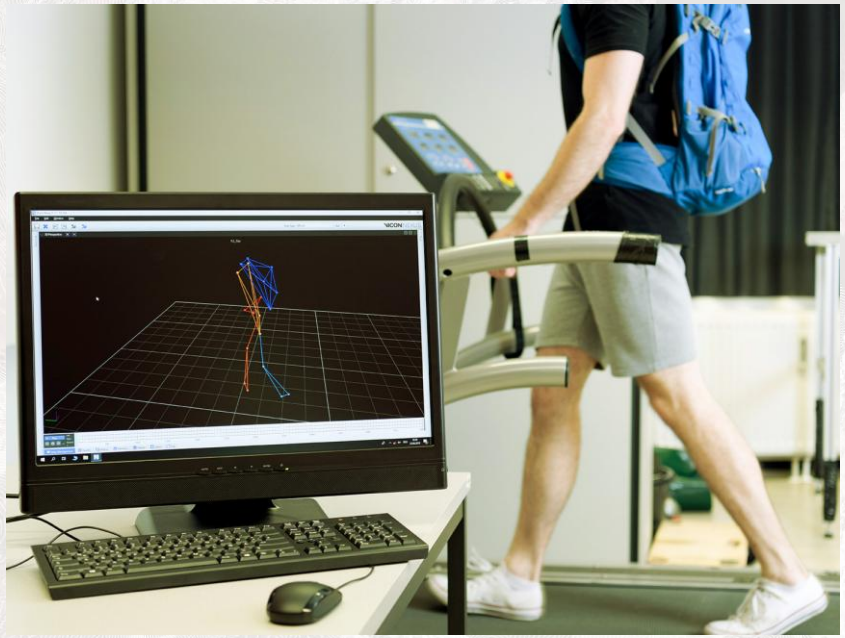
EMG For Core And Back Muscles

3rd Concept

USWE No Dancing Monkey

IMU Sensor Bag Swing Test (Incline Treadmill)

[8]



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Conclusion

Material Substitution

Dimension Adjustments

USWE - No Dancing Monkey



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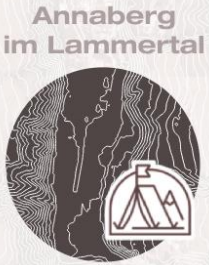
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